



Maths By Gagan Pratap

Trigonometry Sheet-5

Maths Special Batch
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1. In right angled triangle ABC, $\angle B = 90^\circ$, and angle A and angle C are acute angles. If $\operatorname{cosec} A = 2\sqrt{2}$, then find $\sin A \cdot \cos C + \cos A \cdot \sin C$?

समकोण त्रिभुज ABC में, $\angle B = 90^\circ$, और कोण A और कोण C न्यूनकोण हैं। यदि $\operatorname{cosec} A = 2\sqrt{2}$, तो $\sin A \cdot \cos C + \cos A \cdot \sin C$ ज्ञात करें? **SSC CHSL PRE 2024**

- A) 0
B) $2\sqrt{2}$
C) 1
D) $\sqrt{2}$

2. For what value of θ (in degrees) is the following equation true?

निम्न समीकरण में θ (डिग्री में) का सही मान क्या होगा?

$$\sin 3\theta \cos \theta - \cos 3\theta \sin \theta = \frac{1}{2}, 0 < \theta < \frac{\pi}{2}$$

- (a) 45 (b) 30 (c) 60 (d) 15

3. $\frac{\sin 59^\circ \cdot \cos 31^\circ + \cos 59^\circ \cdot \sin 31^\circ}{\cos 20^\circ \cdot \cos 25^\circ - \sin 20^\circ \cdot \sin 25^\circ} = ?$

- (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{3}{\sqrt{2}}$ (c) $\sqrt{3}$ (d) $\sqrt{2}$

4. What is the value of $\cos(x+y) - \sin(x-y) + \tan(2z)$?

$\cos(x+y) - \sin(x-y) + \tan(2z)$ का मान क्या है? **SSC CHSL PRE 2024**

- A) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y - \cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
B) $\cos x \cdot \cos y - \sin x \cdot \sin y - \sin x \cdot \cos y + \cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
C) $\cos x \cdot \cos y - \sin x \cdot \sin y - \cos x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$
D) $\sin x \cdot \sin y + \frac{2 \tan z}{1 - \tan^2 z}$

5. $\frac{\cos 40^\circ - \cos 140^\circ}{\sin 80^\circ + \sin 20^\circ} = ?$

- (a) $\sqrt{\frac{3}{2}}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{\frac{2}{3}}$

6. What is the value of $\frac{\sin(A+B)}{\sin A \cos B}$?

$\frac{\sin(A+B)}{\sin A \cos B}$ का मान क्या है?

- (a) $1 + \cot A \tan B$
(b) $1 + \tan A \cot B$
(c) $1 - \sin A \cos B$
(d) $1 - \cot A \tan B$

7. Given that A and B are second quadrant angles, $\sin A = \frac{1}{3}$ and $\sin B = \frac{1}{5}$, then find the value of $\cos(A-B)$.

दिया गया है कि A और B द्वितीय चतुर्थांश के कोण हैं, $\sin A = \frac{1}{3}$ और $\sin B = \frac{1}{5}$ है, तो $\cos(A-B)$ का मान ज्ञात कीजिए।

- (a) $\frac{4\sqrt{3}+1}{15}$
(b) $\frac{8\sqrt{3}-1}{15}$
(c) $\frac{8\sqrt{3}+1}{15}$
(d) $\frac{4\sqrt{3}+1}{15}$

8. Find the value of $\cos\left(\frac{\pi}{4}-\theta\right) \cdot \cos\left(\frac{\pi}{4}-\phi\right) - \sin\left(\frac{\pi}{4}-\theta\right) \cdot \sin\left(\frac{\pi}{4}-\phi\right)$.

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a) $\sin(\theta-\varphi)$ b) $\sin(\theta+\varphi)$ c) $\cos(\theta-\varphi)$ d) $\cos(\theta+\varphi)$

9. $\cos(36^\circ - A)\cos(36^\circ + A) + \cos(54^\circ + A)\cos(54^\circ - A) = ?$

- A) $\cos 2A$
B) $\cot 2A$
C) $\sin 2A$
D) $\cos 108^\circ$

10. $(\sin x \cos y + \cos x \sin y)(\sin x \cos y - \cos x \sin y) = ?$

a) $\cos^2 y - \cos^2 x$ b) $\cos^2 x - \sin^2 y$ c) $\sin^2 x - \cos^2 y$ d) $\sin^2 y - \sin^2 x$

11. What is the value of the expression $\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$?

$\cos 2A \cos 2B + \sin^2(A - B) - \sin^2(A + B)$ का मान क्या है?

- A) $\sin(2A - 2B)$
B) $\sin(2A + 2B)$
C) $\cos(2A + 2B)$
D) $\cos(2A - 2B)$

12. If $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$, then $x = ?$

यदि $\tan A \tan B + \frac{\cos x}{\cos A \cos B} = 1$ है, तो x का मान क्या होगा?

- A) B
B) A
C) $A + B$
D) $A - B$

13. Simplify the following $\sin 2x + 2\sin 4x + \sin 6x$?

निम्नलिखित $\sin 2x + 2\sin 4x + \sin 6x$ को सरल करें?

- A) $4\cos^2 x \sin 4x$
B) $4\cos^2 x \sin x$
C) $2\cos^2 x \sin 4x$
D) $4\cos^2 x \sin x$

14. $\frac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = ?$

(a) $\tan 2A$ (b) $\tan 8A$
(c) $\cot 4A$ (d) $\tan 4A$

15. Simplify/निम्न का मान ज्ञात कीजिए।

$$\frac{\sin 8\theta \cos \theta - \sin 6\theta \cos 3\theta}{\cos 2\theta \cos \theta - \sin 3\theta \sin 4\theta}$$

(a) $\cot \theta$ (b) $\tan \theta$ (c) $\tan 2\theta$ (d) $\cot 2\theta$

16. $\cos\left[\frac{180-\theta}{2}\right] \cos\left[\frac{180-9\theta}{2}\right] + \sin\left[\frac{180-3\theta}{2}\right] \sin\left[\frac{180-13\theta}{2}\right]$

a) $\sin 2\theta \sin 4\theta$ b) $\cos 2\theta \cos 6\theta$ c) $\sin 2\theta \sin 6\theta$ d) $\cos 2\theta \cos 4\theta$

17. $\frac{\tan 5\theta + \tan 3\theta}{\tan 5\theta - \tan 3\theta} = ?$

- A) $4\cos 2\theta \cos 4\theta$
B) $4\sin 2\theta \cos 4\theta$
C) $4\cos 2\theta \sin 4\theta$
D) $4\sin 2\theta \sin 4\theta$

18. If $\sin A + \sin B = x$ and $\cos A + \cos B = y$, then find $\cos(A - B)$?

यदि $\sin A + \sin B = x$ & $\cos A + \cos B = y$ है, तो $\cos(A - B)$ का मान ज्ञात कीजिये?

- A) $\frac{x^2 + y^2 + 2}{2}$ C) $\frac{x^2 - y^2 + 2}{2}$
B) $\frac{x^2 - y^2 - 2}{2}$ D) $\frac{x^2 + y^2 - 2}{2}$

19. If $\cos x + \cos y = \frac{1}{3}$ & $\sin x + \sin y = \frac{1}{4}$, then find $\sin(x + y)$?

यदि $\cos x + \cos y = \frac{1}{3}$ & $\sin x + \sin y = \frac{1}{4}$ है, तो $\sin(x + y)$ ज्ञात कीजिये ?

- A) $24/25$ C) $16/25$



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B) 32/75

D) 9/16

20. If $\sin\alpha = \frac{4}{5}$ and $\cos\beta = \frac{5}{13}$, then $\cos\frac{\alpha-\beta}{2} = ?$

a) $\frac{8}{\sqrt{63}}$

b) $\frac{63}{65}$

c) $\frac{8}{\sqrt{65}}$

d) $\frac{4}{\sqrt{65}}$

21. $\sin 75^\circ + \sin 15^\circ = ?$

(a) $\sqrt{3}$

(b) $2\sqrt{3}$

(c) $\sqrt{\frac{3}{2}}$

(d) $\frac{3}{\sqrt{2}}$

22. $\tan 15^\circ - \cot 165^\circ = ?$

a) 4

b) $2\sqrt{3}$

c) 2

d) $\sqrt{3}+2$

23. If $\tan 15^\circ = 2 - \sqrt{3}$, then the value of $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$ is

यदि $\tan 15^\circ = 2 - \sqrt{3}$ तो $\tan 15^\circ \cot 75^\circ + \tan 75^\circ \cot 15^\circ$ का मान क्या है?

(a) 6

(b) 8

(c) 14

(d) 10

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24. $\left[\frac{\operatorname{cosec} 31^\circ - \sec 59^\circ}{\sin 18^\circ \div \cos 72^\circ} \times \frac{\tan 26^\circ}{\cot 64^\circ} \right] + \left[\frac{\tan 30^\circ + \tan 15^\circ}{1 - \tan 30^\circ \tan 15^\circ} \right] = ?$

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A) 0

B) -1

C) 1

D) 2

25. Find $\frac{\sin 13^\circ \cdot \sin 77^\circ}{\sin 154^\circ} = ?$

$\frac{\sin 13^\circ \cdot \sin 77^\circ}{\sin 154^\circ}$ ज्ञात कीजिए?

A) 1

B) 0.5

C) 2

D) 0.25

26. Find the value of $\sqrt{\frac{1-\tan A}{1+\tan A}}$

$\sqrt{\frac{1-\tan A}{1+\tan A}}$ का मान ज्ञात कीजिये?

A) $\sqrt{\frac{1+\sin A}{\cos A}}$

B) $\sqrt{\frac{1+\sin 2A}{\cos 2A}}$

C) $\sqrt{\frac{1-\sin A}{\cos A}}$

D) $\sqrt{\frac{1-\sin 2A}{\cos 2A}}$

27. If $\tan 12^\circ = x$, then find the value of $\cos 66^\circ$?

यदि $\tan 12^\circ = x$ है तो $\cos 66^\circ$ का मान ज्ञात कीजिये?

A) $\frac{2x}{1+x^2}$

B) $\frac{x}{1+x^2}$

C) $\frac{2x}{1-x^2}$

D) $\frac{2x}{x^2-1}$

28. If $\tan 16^\circ = a$, then find $\operatorname{cosec} 302^\circ$?

यदि $\tan 16^\circ = a$, तो $\operatorname{cosec} 302^\circ$ ज्ञात कीजिए?

A) $\frac{a^2+1}{a^2-1}$

B) $\frac{a^2}{a^2+1}$

C) $\frac{a^2-1}{a^2+1}$

D) $\frac{1-a^2}{a^2+1}$

29. If $\cos^4 \theta - \sin^4 \theta = \frac{4}{5}$, then find $\sin 4\theta$?

यदि $\cos^4 \theta - \sin^4 \theta = \frac{4}{5}$, तो $\sin 4\theta$ ज्ञात करें? SSC CHSL PRE 2024

A) 24/25

B) 21/25

C) 16/25

D) 18/25



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30. If $\sin x = \frac{2}{3}$, then find the value of $\cos 3x$.

यदि $\sin x = \frac{2}{3}$ है, तो $\cos 3x$ का मान ज्ञात कीजिए।

- (a) 0.6735 (b) -0.8765
(c) -0.5797 (d) 0.5678

31. If $\sin(x) = \frac{2}{5}$ and x is an acute angle, find the exact values of $\cos(4x) - \cos(2x)$.

यदि $\sin(x) = \frac{2}{5}$ और x एक न्यून कोण है, तो $\cos(4x) - \cos(2x)$ का सटीक मान ज्ञात कीजिए।

- (a) $\frac{625}{32}$
(b) $-\frac{472}{625}$
(c) $-\frac{427}{625}$
(d) $-\frac{472}{25}$

32. If $\cos^4 a - \sin^4 a = \frac{5}{6}$, then the value of $2\cos^2 a - 1$?

यदि $\cos^4 a - \sin^4 a = \frac{5}{6}$ है, तो $2\cos^2 a - 1$ का मान ज्ञात करें।

- (a) 6/11
(b) 5/6
(c) 6/5
(d) 11/6

33. The value of $\left(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}\right)^3 \frac{1}{\tan\theta} - \sec^2\theta$ is:

$\left(\frac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta}\right)^3 \frac{1}{\tan\theta} - \sec^2\theta$ का मान ज्ञात कीजिए।

- (a) 2 (b) 1
(c) 0 (d) -1

34. If $\sin(A+B) = \cos(A+B)$, what is the value of $\tan A$?

यदि $\sin(A+B) = \cos(A+B)$, तो $\tan A$ का मान क्या है?

- A) $\frac{1-\tan B}{1+\tan B}$
B) $\frac{1+\tan B}{1-\tan B}$
C) $\frac{1+\sec B}{1-\sec B}$
D) $\frac{1-\csc B}{1+\csc B}$

35. What is the value of $\frac{\cot 60^\circ - \cot 30^\circ}{1 + \cot 60^\circ \cdot \cot 30^\circ}$?

$\frac{\cot 60^\circ - \cot 30^\circ}{1 + \cot 60^\circ \cdot \cot 30^\circ}$ का मान क्या है?

- (a) $1/\sqrt{3}$
(b) $\sqrt{3}$
(c) $-\sqrt{3}$
(d) $-1/\sqrt{3}$

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36. $\frac{3\cos^3\theta - 2\cos\theta}{\sin\theta - 3\sin^3\theta}$ is equal to:

$\frac{3\cos^3\theta - 2\cos\theta}{\sin\theta - 3\sin^3\theta}$ का मान ज्ञात कीजिए।

- [A] $\tan\theta$ [B] $\cot^2\theta$
[C] $\cot\theta$ [D] $\tan^2\theta$

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37. What will be the value of $\sin 10^\circ - 4/3 \sin^3 10^\circ$?

$\sin 10^\circ - 4/3 \sin^3 10^\circ$ का मान क्या होगा?

- (a) $\frac{1}{3\sqrt{3}}$
(b) $\frac{1}{6}$
(c) $\frac{1}{2\sqrt{3}}$
(d) $\frac{3}{6}$

[SSC CGL 2022]

38. What is $\tan \frac{\theta}{2}$?

$\tan \frac{\theta}{2}$ क्या है?

1. $\frac{\cos \theta}{1 - \sin \theta}$

2. $\frac{\sin \theta}{1 - \sin \theta}$

3. $\frac{\cos \theta}{1 + \cos \theta}$

4. $\frac{\sin \theta}{1 + \sin \theta}$

[SSC CGL 2023 PRE]

39. Simplify: $1 - 8\cos^2 \theta + 8\cos^4 \theta$

- (a) $\cos 4\theta$ (b) $\sin 4\theta$ (c) $\cos 6\theta$ (d) $\cos 8\theta$

40. If $\sin^6 \theta + \cos^6 \theta = \sin 2\theta$, then find $\cos 4\theta$?

यदि $\sin^6 \theta + \cos^6 \theta = \sin 2\theta$ है, तो $\cos 4\theta$ ज्ञात कीजिये ?

- A) $\sqrt{5}/3$ C) $1/3$
B) $2/3$ D) $1/9$

41. If $\tan 6\theta = \frac{4}{3}$, then find $\frac{1}{2}(4\operatorname{cosec} 2\theta - 3\sec 2\theta)$?

यदि $\tan 6\theta = \frac{4}{3}$ है, तो $\frac{1}{2}(4\operatorname{cosec} 2\theta - 3\sec 2\theta)$ ज्ञात कीजिये ?

- A) 5 C) $5/2$
B) 10 D) 4

42. Simplify $\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$.

$\frac{\cos 13^\circ + \sin 13^\circ}{\cos 13^\circ - \sin 13^\circ}$ को हल कीजिए।

[SSC CPO 2024]

- [A] $\cot 58^\circ$
[C] $\tan 58^\circ$

- [B] $\tan 32^\circ$
[D] $\cos 26^\circ$

43. $\frac{\cos 11^\circ - \sin 11^\circ}{\cos 11^\circ + \sin 11^\circ} = \cot 4\theta$, then $\theta = ?$

- (a) 11° (b) 14° (c) 9° (d) 16°

44. $\tan 70^\circ = ?$

- (a) $2 \tan 20^\circ + \tan 50^\circ$
(b) $2 \tan 50^\circ + \tan 20^\circ$
(c) $2 \tan 50^\circ \tan 20^\circ$
(d) None

45. $(1 + \tan 8^\circ)(1 + \tan 37^\circ) = ?$

- (a) 1 (b) 2 (c) 3 (d) 4

46. If $(1 + \tan \alpha)(1 + \tan 4\alpha) = 2$, $\alpha \in (0, \frac{\pi}{16})$ then find the value of α .

यदि $(1 + \tan \alpha)(1 + \tan 4\alpha) = 2$, $\alpha \in (0, \frac{\pi}{16})$

है, तो α का मान ज्ञात कीजिए। [RRB NTPC 2021]

[A] $\frac{\pi}{30}$

[B] $\frac{\pi}{20}$

[C] $\frac{\pi}{40}$

[D] $\frac{\pi}{60}$

47. $(1 + \tan 1^\circ)(1 + \tan 2^\circ)(1 + \tan 3^\circ) \dots (1 + \tan 45^\circ) = ?$

- (a) 2^{21} (b) 2^{22} (c) 2^{23} (d) 2^{24}

48. Find the value of $\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ$.

$\tan 72^\circ - \tan 27^\circ - \tan 72^\circ \tan 27^\circ$ का मान ज्ञात कीजिए।



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- (a) -1
(c) -2

- (b) 1
(d) 0

49. Find the value of $(\cot 13^\circ - 1)(\cot 32^\circ - 1)$?
($\cot 13^\circ - 1$)($\cot 32^\circ - 1$) का मान ज्ञात कीजिए?

- A) 1
B) 2
C) 3
D) -2

50. If $A + B = 585^\circ$, then find the value of $\left(\frac{\tan A}{1 - \tan A}\right)\left(\frac{\tan B}{1 - \tan B}\right)$?

यदि $A + B = 585^\circ$ है, तो $\left(\frac{\tan A}{1 - \tan A}\right)\left(\frac{\tan B}{1 - \tan B}\right)$ का मान ज्ञात कीजिये?

- A) 2
B) 4
C) 0.5
D) 1

51. Find the value of $\frac{\tan 35^\circ + \tan 25^\circ + \sqrt{3} \tan 25^\circ \tan 35^\circ}{\tan 19^\circ + \tan 26^\circ + \tan 19^\circ \tan 26^\circ}$?

$\frac{\tan 35^\circ + \tan 25^\circ + \sqrt{3} \tan 25^\circ \tan 35^\circ}{\tan 19^\circ + \tan 26^\circ + \tan 19^\circ \tan 26^\circ}$ का मान ज्ञात कीजिए?

- A) $-\sqrt{3}$
B) 2
C) $\sqrt{3}$
D) 1

52. Find $\tan 19^\circ 40' + \tan 40^\circ 20' + \sqrt{3} \tan 19^\circ 40' \tan 40^\circ 20'$?

$\tan 19^\circ 40' + \tan 40^\circ 20' + \sqrt{3} \tan 19^\circ 40' \tan 40^\circ 20'$ ज्ञात कीजिये?

- A) $\frac{\sqrt{3}}{4}$
B) $\sqrt{3}$
C) $\frac{\sqrt{3}}{2}$
D) 1

53. $\tan \frac{2\pi}{5} - \tan \frac{\pi}{15} - \sqrt{3} \tan \frac{2\pi}{5} \tan \frac{\pi}{15} = ?$

- a) 1
b) $\sqrt{2}$
c) 0
d) $\sqrt{3}$

54. $(\sqrt{3} + \tan 17^\circ)(\sqrt{3} + \tan 13^\circ)(\sqrt{3} + \tan 11^\circ)(\sqrt{3} + \tan 19^\circ) = ?$

- a) 4
b) 16
c) 8
d) 9

55. In a $\triangle ABC$, $\tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2} = ?$

- (a) 0
(b) 1
(c) -1
(d) 2

56. Find the value of $\tan 27^\circ \tan 34^\circ + \tan 34^\circ \tan 29^\circ + \tan 29^\circ \tan 27^\circ$?

$\tan 27^\circ \tan 34^\circ + \tan 34^\circ \tan 29^\circ + \tan 29^\circ \tan 27^\circ$ का मान ज्ञात कीजिए?

- A) 0
B) -1
C) $\sqrt{3}$
D) 1

57. If $\tan 4A \cdot \tan 5A + \tan 5A \cdot \tan 6A + \tan 6A \cdot \tan 4A = 1$,

then find the value of $\frac{\sec 10A}{\csc \frac{15A}{2}}$?

यदि $\tan 4A \cdot \tan 5A + \tan 5A \cdot \tan 6A + \tan 6A \cdot \tan 4A = 1$ है, तो $\frac{\sec 10A}{\csc \frac{15A}{2}}$ का मान ज्ञात कीजिये?

- A) $\sqrt{3}$
B) $2\sqrt{2}$
C) $\sqrt{2}$
D) 2

58. $\cot 38^\circ \cdot \cot 63^\circ + \cot 63^\circ \cdot \cot 79^\circ + \cot 79^\circ \cdot \cot 38^\circ = ?$

- (a) 0
(b) 1
(c) -1
(d) 3

59. In a triangle XYZ, $\tan X = 2$ and $\tan Y = 4$, then what is the value of $\tan Z$?

एक त्रिभुज XYZ में $\tan X = 2$ और $\tan Y = 4$ है, तो $\tan Z$ का मान क्या है?

- A) $\frac{6}{5}$
B) $\frac{6}{7}$
C) $\frac{5}{6}$
D) $\frac{7}{6}$

60. If $0^\circ < A, B < 45^\circ$, $\cos(A+B) = \frac{24}{25}$ and $\sin(A-B) = \frac{15}{17}$, then $\tan 2A$ is:

यदि $0^\circ < A, B < 45^\circ$, $\cos(A+B) = \frac{24}{25}$ और $\sin(A-B) = \frac{15}{17}$ है तो $\tan 2A$ का मान क्या है?

- a) $\frac{213}{4}$
b) 0
c) 1
d) $\frac{416}{87}$



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61. Find the value of $\alpha + \beta$, if $\sin \alpha = \frac{1}{\sqrt{5}}$ and $\cos \beta = \frac{3}{\sqrt{10}}$,

$\sin \alpha = \frac{1}{\sqrt{5}}$ और $\cos \beta = \frac{3}{\sqrt{10}}$, तब $\alpha + \beta$ का मान ज्ञात कीजिए।

RRB NTPC 2021

[A] 60°

[B] 90°

[C] 45°

[D] 30°

62. If $\tan A - \tan B - \tan C = \tan A \tan B \tan C$, what is the value of A in terms of B and C?

यदि $\tan A - \tan B - \tan C = \tan A \tan B \tan C$ है, तो B और C के संदर्भ में A का मान क्या है?

A) $A = B+C$

B) $A = 2B - 2C$

C) $A = B-C$

D) $A = \frac{B-C}{2}$

63. $\tan 13x - \tan 9x - \tan 4x = ?$

a) $\cot 13x \cdot \cot 9x \cdot \cot 4x$ b) $\tan 13x \cdot \tan 9x \cdot \tan 4x$

c) $1 + \tan 4x \cdot \tan 9x$ d) None

64. $\tan\left(\frac{\pi}{4} + A\right) \times \tan\left(\frac{3\pi}{4} + A\right) = ?$

a) 1 b) 0 c) $\cot \frac{A}{2}$ d) -1

65. $\tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right) = ?$

a) $2\sin 2\theta$ b) $2\tan 2\theta$ c) $2\sec 2\theta$ d) $2\cos 2\theta$

66. If $\frac{\cos 34^\circ}{1 + \sin 34^\circ} = \cot(3\theta + 23^\circ)$, then $\theta = ?$ where $(0 < \theta < 90^\circ)$

(a) 12°

(b) 13°

(c) 14°

(d) 15°

67. If $4\tan\theta - 3 = 0$, then the value of $\frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ is:

यदि $4\tan\theta - 3 = 0$ है, तो $\frac{1 - \cos 2\theta}{1 + \cos 2\theta}$ का मान ज्ञात कीजिए।

(a) 1

(b) $7/15$

(c) $4/3$

(d) $9/16$

68. $\left[\frac{1}{1 - \tan\theta}\right] - \left[\frac{1}{1 + \tan\theta}\right] = ?$

a) $\tan \theta$ b) $\cot 2\theta$ c) $\tan 2\theta$ d) $\cot \theta$

69. Find the value of $\left[\frac{4\cos(90-\theta)\sin^3(90+\theta) - 4\sin(90+\theta)\cos^3(90-\theta)}{\cos\left(\frac{180+8\theta}{2}\right)}\right]$.

$\left[\frac{4\cos(90-\theta)\sin^3(90+\theta) - 4\sin(90+\theta)\cos^3(90-\theta)}{\cos\left(\frac{180+8\theta}{2}\right)}\right]$ का मान ज्ञात कीजिए।

(a) -1

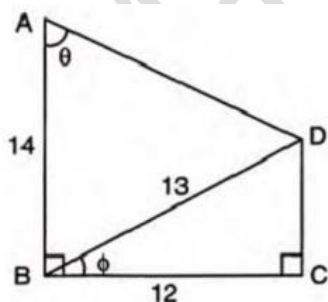
(b) 1

(c) 2

(d) 0

70. Using the measurements given in the following figure, the value of $\frac{\sin\phi + \tan\theta}{\sin\phi - \tan\theta}$ is _____.

निम्नलिखित आकृति में दिए गए मापों का उपयोग करते हुए, $\frac{\sin\phi + \tan\theta}{\sin\phi - \tan\theta}$ का मान है।



(a) $6/35$

(b) $-67/37$

(c) $-7/32$

(d) $7/3$



Maths By Gagan Pratap

Trigonometry Sheet-5

Maths Special Batch
By Gagan Pratap

71. $\frac{\tan 70^\circ - \tan 20^\circ - 2 \tan 40^\circ}{\tan 10^\circ} = ?$

a) 4

b) $2\sqrt{3}$

c) 2

d) 1

72. Find $\cos^2 27^\circ + \cos^2 33^\circ + \cos^2 87^\circ$?

$\cos^2 27^\circ + \cos^2 33^\circ + \cos^2 87^\circ$ ज्ञात कीजिये?

A) 1

B) 2

C) $\frac{1}{2}$

D) $\frac{3}{2}$

73. $\cos^3 10^\circ + \cos^3 110^\circ + \cos^3 130^\circ = ?$

(a) $\frac{3}{4}$

(b) $\frac{3}{8}$

(c) $\frac{3\sqrt{3}}{8}$

(d) $\frac{3\sqrt{3}}{4}$

74. $\tan 10^\circ - \tan 50^\circ + \tan 70^\circ = ?$

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) 3

(d) $\frac{4}{\sqrt{3}}$

75. If $\tan 25^\circ + \tan 85^\circ + \tan 145^\circ = a + b\sqrt{c}$, where $a, b, c \in \mathbb{I}$ and c is prime number, then find $(a+b+c)$?

यदि $\tan 25^\circ + \tan 85^\circ + \tan 145^\circ = a + b\sqrt{c}$, जहाँ $a, b, c \in \mathbb{I}$ और c अभाज्य संख्या है, तो $(a+b+c)$ ज्ञात कीजिए?

A) 12

B) 5

C) 6

D) 10

76. $\cot 20^\circ - \cot 40^\circ + \cot 80^\circ = ?$

(a) $\sqrt{3}$

(b) $\frac{1}{\sqrt{3}}$

(c) 3

(d) $\frac{4}{\sqrt{3}}$

77. The value of $\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 60^\circ \cdot \sin 70^\circ$ is:

$\sin 10^\circ \cdot \sin 50^\circ \cdot \sin 60^\circ \cdot \sin 70^\circ$ का मान क्या है?

(a) $\frac{\sqrt{3}}{16}$

(b) $\frac{\sqrt{3}}{15}$

(c) $\frac{\sqrt{3}}{4}$

(d) $\frac{\sqrt{3}}{2}$

78. $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ = ?$

(a) $\frac{1}{16}$

(b) $\frac{3}{16}$

(c) $\frac{5}{16}$

(d) $\frac{1}{8}$

79. $\tan 20^\circ \cdot \tan 40^\circ \cdot \tan 60^\circ \cdot \tan 80^\circ = ?$

(a) 2

(b) 3

(c) 4

(d) 6

80. $\cos 42^\circ \cdot \cos 78^\circ \cdot \cos 36^\circ = ?$

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{1}{8}$

(d) $\frac{1}{16}$

81. $\sin 12^\circ \cdot \sin 24^\circ \cdot \sin 48^\circ \cdot \sin 84^\circ = ?$

(a) $\frac{5}{16}$

(b) $\frac{1}{8}$

(c) $\frac{1}{16}$

(d) $\frac{3}{16}$

82. $\tan 6^\circ \cdot \tan 42^\circ \cdot \tan 66^\circ \cdot \tan 78^\circ = ?$

(a) 1

(b) $\frac{1}{4}$

(c) $\frac{1}{16}$

(d) $\frac{1}{9}$

83. $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ = ?$

(a) 6

(b) 2

(c) 3

(d) 4

84. If $7 \tan \theta \cdot \tan \phi = 3$, then $\frac{\cos(\theta + \phi)}{\cos(\theta - \phi)} = ?$

(a) $\frac{3}{7}$

(b) $\frac{2}{5}$

(c) $\frac{1}{4}$

(d) $\frac{5}{8}$